

APA NEAGRA, Romania

WMO: 153410

Lat: 44.9969N

Lon: 22.8594E

Elev: 251

StdP: 98.35

Time Zone: 2.00 (EUE)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-10.5	-14.5	1.1	-11.2	-12.0	1.4	-9.0	8.4	6.4	6.4	4.4	0.4	10	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.0	31.9	21.5	30.6	21.3	29.1	21.0	24.4	28.6	23.0	27.8	22.0	26.9	1.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.6	26.3	21.6	16.7	25.2	20.4	15.6	24.2	75.2	27.7	69.5	27.9	65.5	26.9	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.8	4.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-2.0	1.7	5.7	10.9	15.9	19.8	20.8	21.5	16.4	10.3	5.3	0.7
	DBStd	8.74	4.30	4.18	4.18	4.03	2.81	2.76	2.61	2.74	3.67	3.77	4.09	3.77
	HDD10.0	1266	372	233	141	38	1	0	0	0	2	43	147	289
	HDD18.3	3094	630	466	391	224	83	17	9	7	78	249	392	547
	CDD10.0	1491	0	1	8	64	183	293	333	357	193	52	6	0
	CDD18.3	278	0	0	0	0	8	60	84	106	20	0	0	0
	CDH23.3	3269	0	0	1	35	126	659	982	1170	278	20	0	0
	CDH26.7	1047	0	0	0	2	12	186	319	462	65	1	0	0
Wind	WSAvg	1.1	0.7	1.0	1.2	1.6	1.2	1.2	1.3	1.2	1.1	1.1	0.8	0.9
Precipitation	PrecAvg	810	53	47	48	71	81	86	88	47	76	70	66	79
	PrecMax	1363	110	158	112	165	178	158	300	113	146	158	180	210
	PrecMin	344	4	3	2	14	20	12	11	5	4	0	2	2
	PrecStd	224	32	36	34	33	37	45	63	31	48	45	45	62
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	16.0	22.6	26.6	28.2	32.6	33.2	34.3	30.8	25.6	19.0	13.7
		MCWB	7.6	9.4	12.4	16.1	18.7	22.0	22.3	22.6	19.3	15.8	12.9	8.8
	2%	DB	8.8	13.0	19.6	24.1	26.1	30.4	31.4	32.4	28.5	23.1	15.5	11.2
		MCWB	5.6	8.3	11.3	15.8	18.6	21.7	21.5	21.4	19.2	15.2	11.6	7.1
	5%	DB	5.9	10.3	16.2	21.8	24.7	28.8	29.9	31.0	26.5	20.4	13.1	8.6
		MCWB	3.4	6.8	9.6	14.3	18.0	21.3	21.4	21.3	18.2	14.4	10.9	6.0
	10%	DB	3.9	8.2	13.5	19.2	23.1	27.1	28.4	29.2	24.4	17.7	11.1	6.2
		MCWB	2.1	5.7	8.2	12.9	17.4	20.2	20.8	21.2	17.7	13.2	9.8	4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	11.1	13.9	18.0	22.1	25.7	25.8	27.8	23.6	18.3	14.1	9.5
		MCDB	11.4	13.7	21.5	23.5	24.3	28.8	28.6	29.5	28.2	20.1	17.6	11.4
	2%	WB	5.8	9.0	11.9	16.5	20.1	23.2	24.3	24.4	21.4	16.2	12.5	7.7
		MCDB	8.2	11.8	17.8	22.8	23.9	28.3	27.7	29.5	24.5	20.7	14.5	10.7
	5%	WB	4.0	7.5	10.4	15.2	18.7	22.0	22.8	22.8	19.8	15.1	11.4	6.2
		MCDB	5.3	9.4	15.0	21.1	22.8	26.9	27.3	28.1	23.4	19.7	12.9	8.3
	10%	WB	2.5	6.0	9.1	13.5	17.7	21.0	21.7	21.6	18.5	13.7	10.1	4.8
		MCDB	3.5	8.0	12.5	18.1	22.0	26.1	26.7	27.0	22.8	17.2	11.0	6.1
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.0	11.6	12.3	11.6	12.8	13.9	14.0	13.0	12.6	8.6	9.2
		MCDBR	12.4	14.1	17.4	17.2	14.8	15.1	15.7	16.7	16.7	18.3	12.4	13.1
		MCWBR	9.0	9.5	9.6	8.9	7.5	7.0	7.4	7.0	7.4	11.0	8.3	9.4
	5% WB	MCDBR	10.7	12.7	15.0	16.5	12.6	13.5	13.6	14.3	13.1	16.2	10.2	12.4
MCWBR		8.0	9.0	8.9	8.7	7.2	6.9	8.0	7.7	7.1	10.5	7.2	9.2	
Clear-Sky Solar Irradiance	taub	0.340	0.376	0.422	0.491	0.471	0.460	0.478	0.500	0.453	0.408	0.374	0.340	
	taud	2.244	2.147	2.101	1.946	2.064	2.145	2.102	2.032	2.117	2.231	2.267	2.279	
	Ebn at Noon	745	782	797	771	805	815	793	757	757	740	698	702	
	Edh at Noon	87	115	138	176	161	149	154	160	135	105	85	77	
All-Sky Solar Radiation	RadAvg	1.34	2.14	3.31	4.49	5.41	6.08	6.03	5.41	3.85	2.58	1.50	1.12	
	RadStd	0.18	0.29	0.45	0.49	0.47	0.55	0.45	0.51	0.37	0.33	0.20	0.16	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (68 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	-157	+108	+59	

Nomenclature: See separate page